① Python source and its srcML

Python, lines 10–21

```
10 for h in range(hops):
11     a, b = 1 + 2 * h, 2 + 2 * h
12     qc.h(q[a])
13     qc.cx(q[a], q[b])
14     qc.cx(q[src], q[a])
15     qc.h(q[src])
16     qc.measure(q[src], c[2 * h])
17     qc.measure(q[a], c[2 * h + 1])
18     with qc.if_test((c[2 * h + 1], 1)):
19         qc.x(q[b])
20     with qc.if_test((c[2 * h], 1)):
21         qc.z(q[b])
```

srcML, with positions

```
<expr_stmt pos:start="14:5"><expr><call><name><name>qc</name><operator>
    ↪ .</operator><name>cx</name></name><argument_list>(<argument><expr>
    ↪ <name><name>q</name><index>[<expr><name>src</name></expr>]</index>
    ↪ </name></expr></argument>, <argument><expr><name><name>q</name><index>[
    ↪ <expr><name>a</name></expr>]</index></name></expr></argument>)
    ↪ </argument_list></call></expr></expr_stmt>
```

```
<with pos:start="18:5">with <expr><call><name><name>qc</name><operator>
    ↪ .</operator><name>if_test</name></name><argument_list>(<argument><expr>
    ↪ <tuple>(<expr><name><name>c</name><index>[<expr><literal type="number">
    ↪ 2</literal><operator>*</operator><name>h</name><operator>+
    ↪ </operator><literal type="number">1</literal></expr>]</index></name>
    ↪ </expr>, <expr><literal type="number">1</literal></expr>)</tuple>
    ↪ </expr></argument>)</argument_list></call></expr><block>:
    ↪ <block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.
            ↪ </operator><name>x</name></name><argument_list>(<argument>
            ↪ <expr><name><name>q</name><index>[<expr><name>b</name></expr>]
            ↪ </index></name></expr></argument>)</argument_list></call>
            ↪ </expr></expr_stmt>
</block_content></block></with>
```

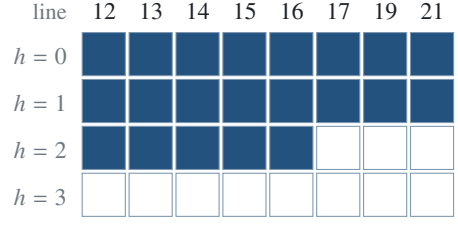
`cx` gate `q[src]` qubit argument, evaluated in each iteration `c[2*h+1]` classical bit, written by a measurement and read by a condition

`if_test` conditioned block `14` source line, from `pos:start`

② Trace of line 8 and iteration $h = 0$

line	h	operation
8	–	u q_0
12	0	h q_1
13	0	cx q_1, q_2
14	0	cx q_0, q_1
15	0	h q_0
16	0	measure $q_0 \to c_0$
17	0	measure $q_1 \to c_1$
19	0	x q_2 if c_1
21	0	z q_2 if c_0
	1–3	the same on the next qubits

③ Slice of `q[4]` by line and iteration



line 8 in the slice, line 23 not **22** of 34 operations